

Minju Hong

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RESEARCH INTERESTS

Theoretical foundations of RLHF; preference learning and pluralistic LLM alignment; optimization dynamics in games; statistical learning theory

AFFILIATION

Carnegie Mellon University

Visiting Researcher, Language Technologies Institute

May. 2026 – Present

Pittsburgh, PA, USA

Korea Advanced Institute of Science and Technology (KAIST)

M.Sc. in Electrical Engineering

Advisor: Prof. Hye Won Chung

Mar. 2024 – Feb. 2026

Daejeon, Korea

Korea Advanced Institute of Science and Technology (KAIST)

B.Sc. in Mathematical Sciences & Electrical Engineering (*Double Major*)

Graduated with Excellence in Leadership and Volunteer Activity

Mar. 2019 – Feb. 2024

Daejeon, Korea

Technical University of Munich

Exchange Student in Mathematics

Apr. 2023 – Sep. 2023

Munich, Germany

PUBLICATIONS

[1] [Provably Efficient Regularized Online RLHF with Generalized Bilinear Preferences](#)

Junghyun Lee, **Minju Hong**, Kwang-Sung Jun, Chulhee Yun, Se-Young Yun.

ICML 2026 Pluralistic Alignment Workshop

CKAIA 2026

SELECTED RESEARCH PROJECTS

Modeling Subjective Annotations in LLM Alignment

Research Intern, Carnegie Mellon University, Language Technologies Institute

Supervised by Prof. Maarten Sap; mentored by Dr. Vasudha Varadarajan

May 2026 – Present

Pittsburgh, PA

- Investigating methods for representing subjective human annotations using probabilistic methods.

Implicit Bias of Softmax Dynamics in Bilinear Zero-Sum Games

Research collaboration with the OptiML Lab, KAIST AI; with Prof. Chulhee Yun

Mar. 2026 – Present

Seoul, Korea

- Characterizing the learning dynamics and implicit regularization of policies.

Regularized Online RLHF with Generalized Bilinear Preferences

M.S. thesis research and continued collaboration

Sep. 2025 – May. 2026

Daejeon, Korea

- Studied online RLHF under generalized bilinear preference models that capture pairwise interactions beyond scalar reward models.
- Contributed to regret analyses for regularized self-play algorithms and statistical complexity analyses for bilinear preference learning.
- Coauthored a paper presented at the ICML 2026 Workshop on Pluralistic Alignment.

RESEARCH EXPERIENCE

Graduate Research Assistant

Advised by Prof. Hye Won Chung

Mar. 2024 - Feb. 2026

Daejeon, Korea

- Investigated the theoretical foundations of order sensitivity in In-Context Learning (ICL) in Large Language Models.
- Analyzed the causal mechanisms behind order sensitivity from an optimization perspective.

Individual Study: Information Spread Dynamics

Undergraduate Researcher, Advised by Prof. Debarghya Ghoshdastidar

Apr. 2023 - Sep. 2023

Munich, Germany

- Studied misinformation mitigation strategies within the Theoretical Foundations of AI Group.
- Developed and evaluated blocker selection algorithms to minimize the spread of misinformation on graph networks.

Project: Mathematical Biology (MAS467)

Sep. 2023 - Dec. 2023

Course Term Project, Advised by Prof. Jaekyung Kim

Daejeon, Korea

Title: *Shedding Light on Combined Multiple Transcriptional Repression Mechanisms*

- Modeled cell cycle dynamics to analyze the stability of transcriptional repression mechanisms.
- Conducted simulations to observe the interaction between multiple repression pathways.

Individual Study: Random Graph Matching

Mar. 2021 - Dec. 2022

Undergraduate Researcher, Advised by Prof. Hye Won Chung

Daejeon, Korea

- Conducted independent study on random graph theory and delivered literature review presentations.
- Focusing on random graph matching algorithms, investigated better algorithmic strategies for stochastic block models.

HONORS AND AWARDS

Doo-Myeong Undergraduate Scholarship

Spring 2020

Awarded to one student from the Department of Mathematics, Class of 2023

A merit-based scholarship granted by the Miwon Group.

National Science and Engineering Undergraduate Scholarship

Spring 2019

Merit-based scholarship awarded by Korea Student Aid Foundation.

Leadership Mileage Certificate, KAIST

Fall 2020

Honor for top 3% among 3,600+ students with top achievements in leadership activities including volunteering and campus activities.

Grand Prize, World Friends Korea ICT Volunteers Award

Fall 2019

Won by the KAIST-Tanzania team for outstanding ICT volunteer projects.

TEACHING EXPERIENCE

Machine Learning Basics and Practices (EE214)

Fall 2025

Teaching Assistant, KAIST

Probability and Introductory Random Processes (EE210)

Spring 2025

Teaching Assistant, KAIST

EE Career Design (EE215)

Spring and Fall 2025

Teaching Assistant, KAIST; provided simultaneous Korean–English interpretation for an international course.

SERVICE AND LEADERSHIP

Graduate Student Representative, KAIST School of Electrical Engineering

Aug. 2024 – Feb. 2025

Organized academic talks and community events for more than 400 graduate students.

Student Council Representative, KAIST Department of Mathematical Sciences

Mar. 2021 – Feb. 2023

Organized academic events and seminars for more than 100 undergraduate students.

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, $\text{\LaTeX}$

LANGUAGES

English Highly proficient (iBT TOEFL 117/120 : R30/L30/S30/W27)
Korean Native